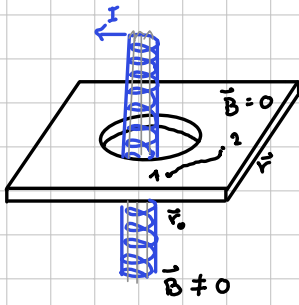


Ahanomav - Bohmov pojav

"Ne smes se poudi
navzajst okrog teh elektr-j."



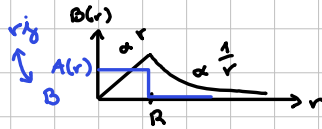
$$\vec{\nabla} \cdot \vec{B} = \vec{\nabla} \times \vec{A} = 0$$

$$\vec{A} = \vec{\nabla} \Lambda \neq 0$$

$$\Lambda(\vec{r}) = \Lambda(\vec{r}_0) - \int_{\vec{r}_0}^{\vec{r}} \vec{A}(\vec{r}') d\vec{r}'$$

analogija:

$$\vec{\nabla} \times \vec{B} = \vec{j}(\mu_0), \quad \vec{B} = \frac{\mu_0 I}{2\pi r} \vec{e}_\varphi, \quad I = \int \vec{j} d\vec{s}$$



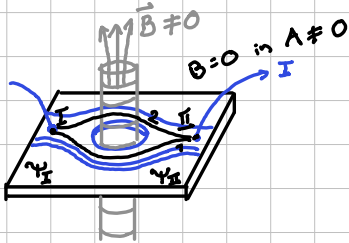
$$i\hbar \frac{\partial \psi_A}{\partial t} = \frac{(\vec{p} - e\vec{A})^2}{2m} \psi_A + V \psi_A$$

$$\vec{B} = 0, \quad \vec{A} \neq 0$$

$$\vec{B} = 0, \quad \vec{A} = 0 : \quad i\hbar \frac{\partial \psi_0}{\partial t} = \frac{p^2}{2m} \psi_0 + V \psi_0$$

$$\vec{A}' = \vec{A} + \vec{\nabla}(-\Lambda) = 0$$

$$\psi_A(\vec{r}, t) = e^{i \frac{e}{\hbar} \int_{\vec{r}_0}^{\vec{r}} \vec{A}(\vec{r}') d\vec{r}'} \psi_0(\vec{r}, t); \quad \vec{B} = 0$$



$$B = 0 : \quad \psi_{II,0} = \psi_1 + \psi_2$$

$$A = 0$$

$$B \neq 0 \quad \vee \quad \text{cilindru}$$

$$A \neq 0 \quad \vee \quad \text{cilindru in zunaj}$$

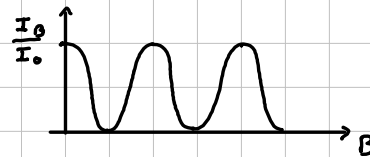
$$\psi_{II,0} = e^{i\delta_1} \psi_1 + e^{i\delta_2} \psi_2$$

$$\delta_1 = \frac{e}{\hbar} \int_I \vec{A}(\vec{r}) d\vec{r}$$

$$\psi_{II,0} = e^{i\delta_2} (e^{i(\delta_1 - \delta_2)} \psi_1 + \psi_2) = e^{i\delta_2} (1 + e^{i(\delta_1 - \delta_2)}) \psi_1$$

$$\delta = \delta_1 - \delta_2 = \frac{e}{\hbar} \left[\int_{I_{pot1}} \vec{A} d\vec{r} + \int_{I_{pot2}} \vec{A} d\vec{r} \right] = \frac{e}{\hbar} \oint \vec{A} d\vec{r} = \frac{e}{\hbar} \int \vec{\nabla} \times \vec{A} \cdot d\vec{s} = \frac{e}{\hbar} \int \vec{B} d\vec{s} = \frac{e}{\hbar} \Phi_B$$

$$\frac{I_0}{I_0} = \frac{|\psi_{II,0}|^2}{|\psi_{I,0}|^2} = \frac{1}{4} |1 + e^{i \frac{e}{\hbar} \Phi_B}|^2 = \cos^2 \frac{e}{2\hbar} \Phi_B$$



maksimum: $e^{i \frac{e}{\hbar} \Phi_B} = 1 \Rightarrow \frac{e}{\hbar} \Phi_B = 2\pi n \rightarrow \Phi_B = \frac{2\pi\hbar}{e} n = \frac{h}{e} n = \Phi_0 n$

$$\Phi = \Phi_0 n; \quad n = 1, 2, 3, \dots \quad n \in \mathbb{Z}$$

$$\Phi_0 = \frac{h}{e} \rightarrow \text{superpredrtilc } e = 2e_0, \quad \Phi_{kvadrat} = \frac{h}{2e_0}$$

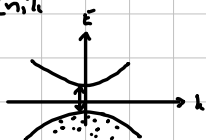
Neobvezna dodatna snov: superprevodnik

3 pomembne lastnosti:

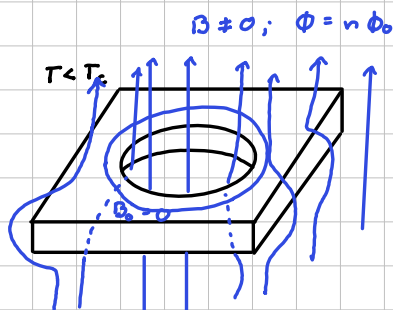
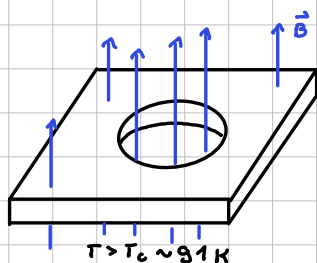
1.) $B=0 \rightarrow$ Meissnerjev pojav $\rightarrow$ idealni diamagnet

2.) $\rho=0 \rightarrow$ idealni prevodnik

3.) energijska vrzel Δ :

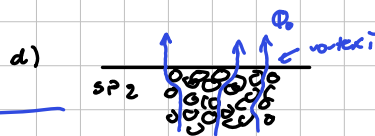
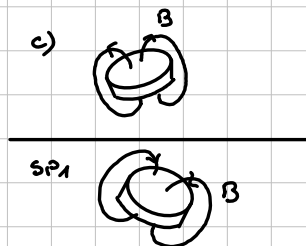
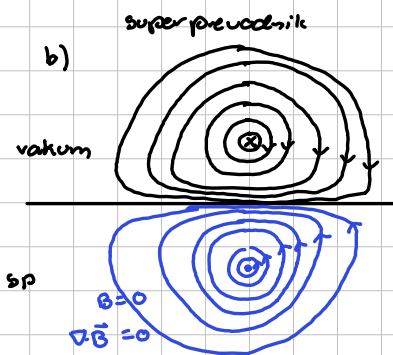
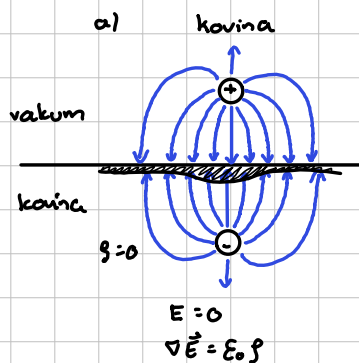


Levitacija:



$$\psi_A = e^{i \frac{e}{\hbar} \int \vec{A} \cdot d\vec{r}}$$

$$\psi_2 = \psi_1 e^{i\phi} \leftarrow \int \vec{A} \cdot d\vec{r} = \int \nabla \vec{A} \cdot d\vec{s} = \Phi = n\Phi_0$$



ko zavreješ magnet
zavreješ vorteks vrti (pinning)

umesto zrak / druga kovina $\rightarrow$ vorteksi pletenje
in so zaljubljeni vrti.

Spin

$$[S_x, S_y] = i\hbar \epsilon_{\alpha\beta\gamma} S_\gamma$$

$$\vec{S} = (S_x, S_y, S_z) \quad , \quad S_\pm = S_x \pm iS_y \quad , \quad S_z |s, m\rangle = m\hbar |s, m\rangle$$

$$S^2 |s, m\rangle = s(s+1)\hbar^2 |s, m\rangle \quad , \quad s = 0, \frac{1}{2}, 1, \frac{3}{2}$$

$$S_\pm |s, m\rangle = \hbar \sqrt{s(s+1) - m(m\pm 1)} |s, m\pm 1\rangle \quad , \quad -s \leq m \leq s$$

$$s = \frac{1}{2} \quad , \quad m = \pm \frac{1}{2}$$

$$|s, \frac{1}{2}\rangle = |\frac{1}{2}, \frac{1}{2}\rangle$$

$$|s, -\frac{1}{2}\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle$$

osnova za kvantno računalništvo
↓
kubit

Uporabljamo zapis: $|m\rangle, |\uparrow\rangle, |\downarrow\rangle$

$$|m\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle \quad , \quad \alpha^2 + \beta^2 = 1 \quad , \quad |\uparrow\rangle \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$X_\uparrow = \begin{pmatrix} 1 & \\ & 0 \end{pmatrix} \quad , \quad X_\downarrow = \begin{pmatrix} 0 & \\ & 1 \end{pmatrix} \quad , \quad X = \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix} = \alpha X_\uparrow + \beta X_\downarrow$$

iz $|\downarrow\rangle$ naredi $|\uparrow\rangle$

$$S_+ |\downarrow\rangle = \hbar \sqrt{\frac{1}{2}(\frac{1}{2}+1) - (-\frac{1}{2})(-\frac{1}{2}+1)} |\uparrow\rangle = \hbar |\uparrow\rangle$$

iz $|\uparrow\rangle$ naredi 0

$$S_+ |\uparrow\rangle = 0$$

$$S_- |\downarrow\rangle = 0 \quad , \quad S_- |\uparrow\rangle = \hbar |\downarrow\rangle$$

$$\Rightarrow S_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} (S_+)_{\uparrow\uparrow} & (S_+)_{\uparrow\downarrow} \\ (S_+)_{\downarrow\uparrow} & (S_+)_{\downarrow\downarrow} \end{pmatrix}$$

$$S_- = (S_+)^\dagger = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Od prej za $l=1$:

$$S_+ = \hbar \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}$$

$$S_x = \frac{1}{2}(S_+ + S_-) = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = S_x^\dagger$$

$$S_y = \frac{1}{2i}(S_+ - S_-) = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = S_y^\dagger$$

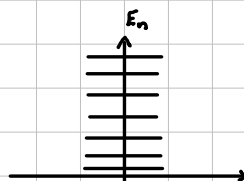
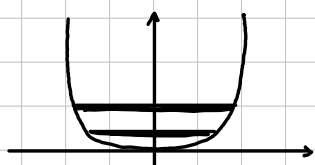
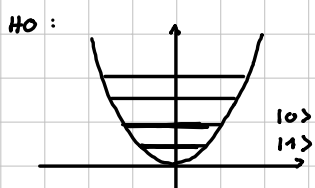
$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

"pri, ki ga učiš
o $\odot$ "
↓

Paulijeve matrice: $\vec{S} = \frac{\hbar}{2} \vec{\sigma} \quad , \quad \vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z) \quad , \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad , \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad , \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Primer uporabe matric: dvonivojski sistem:



baza: $|1\rangle, |2\rangle$

$$H = c_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + c_4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} =$$

$$= E_0 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + E_1 \sigma_x + E_2 \sigma_y + E_3 \sigma_z = \boxed{E_0 I + \vec{E} \vec{\sigma} = H} \quad E_i \in \mathbb{R}$$

Lastnosti:

- $\det \sigma_\alpha = -1$

- $\text{tr } \sigma_\alpha = 0$

- $\sigma_x^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I (=1)$, $\sigma_y^2 = \sigma_z^2 = 1$

$$\sigma_x = \sigma_y = \sigma_z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ 1 & 0 \end{pmatrix} \sigma_z = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \sigma_z = i \sigma_z^2 = i I = i \sigma_x^2 = i \sigma_y^2 = i \sigma_z^2$$

- $\sigma_\alpha \sigma_\beta = i \sigma_\gamma$ $\begin{matrix} \alpha=x & \beta=y \\ \gamma=z \end{matrix}$ $\hookrightarrow xy \rightarrow z$

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = -i \sigma_x \sigma_y \sigma_z$$

$$\sigma_y \sigma_x = -i \sigma_z = -\sigma_x \sigma_y$$

$$\sigma_\alpha \sigma_\beta = \delta_{\alpha\beta} I + i \epsilon_{\alpha\beta\gamma} \sigma_\gamma$$

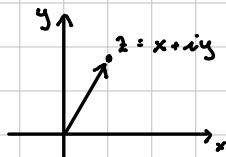
- $[\sigma_\alpha, \sigma_\beta] = 2i \epsilon_{\alpha\beta\gamma} \sigma_\gamma \leadsto \begin{matrix} \vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z) \\ \vec{a} = (a_x, a_y, a_z) \end{matrix}$

$$\vec{a} \cdot \vec{\sigma} = \sum_\alpha a_\alpha \sigma_\alpha = a_x \sigma_x + a_y \sigma_y + a_z \sigma_z$$

- $(\vec{a} \cdot \vec{\sigma})(\vec{b} \cdot \vec{\sigma}) = \sum_{\alpha\beta} a_\alpha \overset{\alpha\beta \rightarrow \gamma}{\sigma_\alpha \sigma_\beta} b_\beta = \sum_\alpha a_\alpha b_\alpha I + i \sum_{\alpha\beta\gamma} \epsilon_{\alpha\beta\gamma} \sigma_\gamma a_\alpha b_\beta = \vec{a} \cdot \vec{b} I + i (\vec{a} \times \vec{b}) \cdot \vec{\sigma}$

Kvaternioni → škola v 4D

Hamilton, 1801 Dublin



Gauss

$$i^2 = j^2 = k^2 = ijk = -1$$

$$q = a_0 + a_1 i + a_2 j + a_3 k, \quad a_\alpha \in \mathbb{R}, \mathbb{C}, \quad q \in \mathbb{H}$$

3 imaginarne konstante → Paulijev matrice

$$ii = -1, \quad i^2 = -1, \quad ij = -k$$

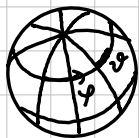
$$i \rightarrow -i\sigma_x$$

$$j \rightarrow -i\sigma_y$$

$$k \rightarrow -i\sigma_z$$

Cayley → vpelje matrice 1853

→ tu se niso bile upejane matrice



(φ, ϑ)

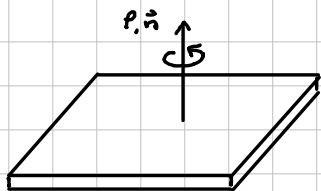
$$\varphi \rightarrow \varphi + \pi$$



→ problem za navigacijo → poljubna napaka → odpravi se s kvaternioni

Rotacija spinorjev

spinor: $\chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|\uparrow\rangle$, $|m\rangle$



zavrtimo stanje χ za $\varphi = \varphi \hat{n}$

$$U = U(\varphi \hat{n}) = e^{-i\varphi \frac{\hat{n} \cdot \vec{\sigma}}{2}}$$

$$U|\chi\rangle = |\tilde{\chi}\rangle$$

$$e^{-i\varphi \frac{\hat{n} \cdot \vec{\sigma}}{2}} = e^{-i\frac{\varphi}{2} \hat{n} \cdot \vec{\sigma}} \quad \text{kot neka transformacija za polovične kote}$$

$$= \sum_{k=0}^{\infty} \frac{1}{k!} \left(-i\frac{\varphi}{2} \hat{n} \cdot \vec{\sigma}\right)^k = 1 - i\frac{\varphi}{2} \hat{n} \cdot \vec{\sigma} + (-i\frac{\varphi}{2})^2 \frac{1}{2!} \underbrace{(\hat{n} \cdot \vec{\sigma})^2}_1 + \dots$$

$$(\hat{n} \cdot \vec{\sigma})^2 = (\hat{n} \cdot \vec{\sigma})(\hat{n} \cdot \vec{\sigma}) = \hat{n} \cdot \hat{n} + i(\hat{n} \times \hat{n}) \cdot \vec{\sigma} = 1$$

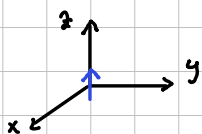
$$(\hat{n} \cdot \vec{\sigma})^3 = 1 \hat{n} \cdot \vec{\sigma}$$

$$\Rightarrow (\hat{n} \cdot \vec{\sigma})^k = \begin{cases} I & k \text{ sodo} \\ \hat{n} \cdot \vec{\sigma} & k \text{ liho} \end{cases}$$

$$(\hat{n} \cdot \vec{\sigma})^4 = ((\hat{n} \cdot \vec{\sigma})^2)^2 = 1$$

$$\boxed{e^{-i\frac{\varphi}{2} \hat{n} \cdot \vec{\sigma}} = \left(1 - \frac{1}{2!} \left(\frac{\varphi}{2}\right)^2 + \frac{1}{4!} \left(\frac{\varphi}{2}\right)^4 - \dots\right) I + (-i) \left(\frac{\varphi}{2} - \frac{1}{3!} \left(\frac{\varphi}{2}\right)^3 + \frac{1}{5!} \left(\frac{\varphi}{2}\right)^5 - \dots\right) \hat{n} \cdot \vec{\sigma} = \cos \frac{\varphi}{2} I - i \sin \frac{\varphi}{2} \hat{n} \cdot \vec{\sigma}}$$

Primer:



$$\chi_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \chi_z$$

$$\sigma_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\sigma_y \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \lambda \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\chi_y = U(\varphi \hat{n}) \chi_z, \quad \hat{n} = (0, 1, 0), \quad \hat{n} \cdot \vec{\sigma} = \sigma_y, \quad \varphi = -\pi/2$$

$$U = \begin{pmatrix} \cos \varphi/2 & -i \sin \varphi/2 \\ i \sin \varphi/2 & \cos \varphi/2 \end{pmatrix} = \begin{pmatrix} \cos \pi/4 & i \sin \pi/4 \\ -i \sin \pi/4 & \cos \pi/4 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & i \frac{1}{\sqrt{2}} \\ i \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

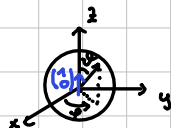
ker vrtno za negativen kot

$$\chi_y = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ i \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}; \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

"Veško leto na arketah dobimo zelo slabe orene pri spodbujanju samostojnega razmišljanja. No, da vas spodbujam: kako bi vi to naredili? (podate sekundo) No pa ker vas nimam preveč časa spodbujati ..."

Splošna rotacija

Blochova sfera:



$$\vec{n}_0 = (\cos\varphi \sin\vartheta, \sin\varphi \sin\vartheta, \cos\vartheta)$$

$$\chi = U \begin{pmatrix} 1 \\ 0 \end{pmatrix} = U(\varphi \vec{e}_2) U(\vartheta \vec{e}_y) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^{-i\frac{\vartheta}{2}\sigma_z} \left(\cos\frac{\vartheta}{2} I - i \sin\frac{\vartheta}{2} \begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix} \right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

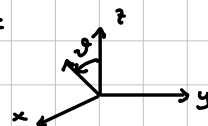
$$= e^{-i\frac{\vartheta}{2}\sigma_z} \begin{pmatrix} \cos\frac{\vartheta}{2} & -\sin\frac{\vartheta}{2} \\ \sin\frac{\vartheta}{2} & \cos\frac{\vartheta}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^{-i\frac{\vartheta}{2}\sigma_z} \begin{pmatrix} \cos\frac{\vartheta}{2} \\ \sin\frac{\vartheta}{2} \end{pmatrix}$$

$$= e^{-i\frac{\vartheta}{2}\sigma_z} \left(\cos\frac{\vartheta}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sin\frac{\vartheta}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) = \left(e^{-i\frac{\vartheta}{2} \cdot 1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos\frac{\vartheta}{2} + \sin\frac{\vartheta}{2} e^{-i\frac{\vartheta}{2}(-1)} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) =$$

$$= \underbrace{e^{-i\frac{\vartheta}{2}}}_{e^{i\varphi}} \left(\cos\frac{\vartheta}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + e^{i\varphi} \sin\frac{\vartheta}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

$$\chi_{\varphi\vartheta} = U(\varphi, \vartheta) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^{-i\varphi\sigma_z} \begin{pmatrix} \cos\frac{\vartheta}{2} \\ e^{i\varphi} \sin\frac{\vartheta}{2} \end{pmatrix}$$

splošna oblika spinorja
 $(\vec{n}_0 \cdot \vec{\sigma}) \chi_{\varphi\vartheta} = \frac{\hbar}{2} \chi_{\varphi\vartheta}$

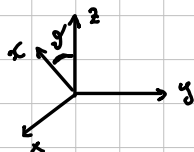


Rotacija spinorjev

vaja:

$$\langle \chi | \sigma_x | \chi \rangle = \left(\cos\frac{\vartheta}{2}, e^{-i\varphi} \sin\frac{\vartheta}{2} \right) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos\frac{\vartheta}{2} \\ \sin\frac{\vartheta}{2} \end{pmatrix} =$$

$$= \left(\cos\frac{\vartheta}{2}, e^{-i\varphi} \sin\frac{\vartheta}{2} \right) \begin{pmatrix} e^{i\varphi} \sin\frac{\vartheta}{2} \\ \cos\frac{\vartheta}{2} \end{pmatrix} = \cos\frac{\vartheta}{2} \sin\frac{\vartheta}{2} (e^{i\varphi} + e^{-i\varphi}) \frac{2}{2} = \cos\varphi \sin\vartheta$$



$$\langle \sigma_y \rangle = \sin\varphi \sin\vartheta$$

$$\langle \sigma_z \rangle = \cos\vartheta$$

Sklopitev delca s spinom $\neq \vec{E}$ in $\vec{B}$

a) $H = -\vec{\mu} \cdot \vec{B}$, $\vec{\mu} = g \frac{e}{2m} \vec{S}$, $g = 2.002 \dots$ Dirac

$$H = -\frac{e}{m} \vec{S} \cdot \vec{B}$$

b) $\vec{B} = -\frac{1}{c^2} (\vec{v} \times \vec{E})$, $\vec{E} = -\nabla \overbrace{\frac{1}{c} V(r)}^{\Phi(r)} = \frac{1}{c} \frac{\partial V(r)}{\partial r} \frac{\vec{r}}{r}$

$$\vec{B} = -\frac{1}{mc^2} \left(\underbrace{m\vec{v}}_{\vec{p}} \times \frac{\partial V}{\partial r} \frac{\vec{r}}{r} \right) = \frac{1}{emc^2 r} \frac{\partial V}{\partial r} \vec{L}$$

$$\Rightarrow H_{LS} = \underbrace{\alpha \frac{\partial V(r)}{\partial r}}_{\downarrow} \vec{L} \cdot \vec{S}$$

c) $H_{\text{Rashba}} = \alpha (\vec{\sigma} \times \vec{p}) \cdot \vec{n}$

Paulijev formalizem

$$a) \quad \Psi(\vec{r}, t) = \begin{pmatrix} \psi_{\uparrow}(\vec{r}, t) \\ \psi_{\downarrow}(\vec{r}, t) \end{pmatrix}$$

izvaž za homogeno polje

$$b) \quad H = \left(\frac{p^2}{2m} + V(\vec{r}, t) - \frac{e}{2m} \vec{L} \cdot \vec{B} \right) I - \frac{ge}{2m} \vec{S} \cdot \vec{B} + W(r) \vec{L} \vec{S} + \alpha (\vec{\sigma} \times \vec{p}) \cdot \vec{n}$$

$$H = \begin{pmatrix} H_{\uparrow\uparrow} & H_{\uparrow\downarrow} \\ H_{\downarrow\uparrow} & H_{\downarrow\downarrow} \end{pmatrix}$$

